

SEEMAB — CLASS 9 (FEDERAL BOARD)

Physics exam-16 — Detailed Marking Report

Units 1-6 (to Example 6.4) and 9 • 38 marks, no choices • attempted and marked 16 September 2026 • independently deep-checked

36.5/38

TOTAL

96.1%

PERCENTAGE

A+

GRADE

10/10

SECTION A

11/12

SECTION B

15.5/16

SECTION C

The short version. Seemab scored 36.5 out of 38, her second-best paper of the year after Physics exam-15. The MCQs were perfect, and more importantly they were perfect in the way that fixes yesterday's two Maths slips: every answer numbered as on the paper, all ten present, and every letter matching her own working. Both long numericals were essentially complete. The 1.5 marks she lost are three half-marks for small recording slips: a wrong unit, one wrongly copied value, and one missing concluding sentence.

SECTION A — Multiple Choice (10 / 10)

Q	Topic	Hers	Key	Her working or reason
1	Acceleration from speeds (gradient)	A	A	$(20 - 5)/5 = 3 \text{ m/s}^2$. Gradient re-test held (wrong on exam-12 and exam-14, right on exam-15 and again here).
2	Momentum	B	B	$4 \times 7 = 28 \text{ kg}\cdot\text{m/s}$, with the unit.
3	Newton's third law	D	D	A written reason that eliminates each wrong option in turn.
4	Centre of gravity	C	C	"other options have different definitions".
5	Second condition of equilibrium	A	A	"the condition clearly said that it must be zero".
6	Liquid pressure	C	C	$1000 \times 10 \times 4 = 40,000 \text{ Pa}$.
7	Torque	B	B	$20 \times 0.3 = "6\text{N}"$. Right letter; the unit of torque is N m, not N (no mark attached in an MCQ).
8	Power	D	D	$1800 \div 6 = 300 \text{ W}$.
9	Wind turbine	A	A	"other options aren't true".
10	Biophysics	B	B	"because the statement said diagnosis, treatment, respiration so it connects to biology".

Both of yesterday's MCQ slips are fixed

On Maths exam-18 she skipped a question because her answer numbering drifted, and on another she worked out option C and wrote D. Today's paper told her in the instructions to number answers as on the paper, count all ten, and check the letter against her working. **She did all three.** Her MCQ answers run 1 to 10 in order, and on every question with working (1, 2, 6, 7, 8) the letter matches the value she computed. She also placed her MCQs after Sections B and C, which is fine.

SECTION B — Short Questions (11 / 12)

Q.2(i)(a) Magnitude of the resultant of 9 N and 12 N 1.5 / 2

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magnitude:- sqrt((9)^2 + (12)^2)
= sqrt(81 + 144)
= sqrt225
15 m
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Method and value are right. The unit is written as **m** (metres), but the resultant of two forces is a force, so it is **15 N**. Half a mark off the answer.

Q.2(i)(b) Direction from the 9 N force 2 / 2

$\tan\theta = 12/9 = 4/3$, " $\theta \approx 53.1^\circ$ from the 9N force." Exactly as asked, including the reference force.

Q.2(ii) Metre rule balance 4 / 4

(a) “Clockwise moment = Anticlockwise moment”, $6 \times 20 = W \times 15 \text{ cm}$, $120 \text{ N cm} / 15 \text{ cm} = 8 \text{ N}$. (b) “Principle of moments.” Her book (Unit 4, §4.4.2) says the second condition of equilibrium “is also called the principle of moments”, so this names the condition and earns the mark.

Teaching point, no mark lost: she labelled the 6 N weight’s moment as clockwise, but a weight hanging to the *left* of the pivot turns the rule **anticlockwise**. The equation is the same either way, so the answer is unaffected.

Q.2(iii)(a) Work done against gravity 1.5 / 2

$$\begin{aligned} m &= 50 \text{ kg} & h &= 4 \text{ m} & g &= 10 \text{ m/s}^2 & t &= 8 \text{ s} \\ \text{Work} &= mgh \\ &= 50 \times 10 \times 8 \\ &= 2000 \text{ J} \end{aligned}$$

She listed $h = 4 \text{ m}$ correctly, then substituted **8** (the time) in place of the height. $50 \times 10 \times 8$ would be 4000, yet her answer **2000 J** is the correct one, so the slip is in copying the value into the line, not in the physics. Half of the substitution mark off; the answer mark stands.

Q.2(iii)(b) Power 2 / 2

“ $P = \text{work} \div \text{time}$ ”, $2000/8 = 250 \text{ W}$, unit included.

SECTION C — Long Questions (15.5 / 16)**Q.3(i) Car of mass 600 kg, rest to 20 m/s in 10 s** 8 / 8

(a) $a = (v_f - v_i)/t = 2 \text{ m/s}^2$. (b) $S = v_i t + \frac{1}{2} a t^2 = 100 \text{ m}$. (c) $F = ma = 600 \times 2 = 1200 \text{ N}$. (d) $\Delta p = mv_f - mv_i = 12000 \text{ kg}\cdot\text{m/s}$, **with the unit the question asked for**. Every part shows formula, substitution and answer with unit: exactly what instruction 5 asked.

Q.3(ii) Hydraulic jack 7.5 / 8

(a) $P = F_1/A_1 = 100/0.004 = 25,000 \text{ Pa}$ (2/2). (b) $F_2 = P \times A_2 = 25,000 \times 0.2 = 5000 \text{ N}$ (2/2).

(c) **1.5/2**. She wrote $F_2/F_1 = A_2/A_1$, $5000/100 = 0.2/0.004$, “ $50 = 50$ ”, and stopped. The question said *show that* the force is multiplied by the same number of times as the area ratio, so one sentence was needed: “Hence the force is multiplied 50 times, the same as the ratio of the areas.” This is the same half-mark as Maths exam-17 3(ii)(b), where “ $5 + 5 = 10$ ” had no “hence collinear”.

(d) **2/2**, in her words: “liquids are nearly incompressible, so the pressure applied to the small piston is transmitted effectively to the large piston. A gas is compressible, so it would compress instead of transmitting the pressure effectively.” A clear, complete explanation.

WHAT TO REVISE

Item	Why
Units match the quantity	A resultant force is in N, not m (2(i)(a)); torque is N m, not N (MCQ 7).
Copy the right value into the formula	$h = 4 \text{ m}$ was listed, then 8 (the time) was substituted (2(iii)(a)).
“Show that” ends with a sentence	“ $50 = 50$ ” needs “hence the force is multiplied 50 times, the same as the area ratio” (3(ii)(c)).
Clockwise or anticlockwise	A weight on the left of a pivot turns anticlockwise (2(ii), no mark lost).

Marking method. First pass on all 8 pages, with the three deciding items (the unit in 2(i)(a), the substituted value in 2(iii)(a), and MCQ 7’s unit) re-read at 300 DPI. An independent re-verification then re-derived all ten MCQ answers from the printed paper (no key errors), ran a shift guard (offset 0 confirmed by her own working on five questions), checked her textbook for the principle-of-moments wording, and re-marked every written part. It agreed with the first pass on every mark. Calculator use is allowed and was not considered.

— END OF MARKING REPORT —